VAMBA: EFFICIENT HOUR-LONG VIDEO UNDERSTANDING

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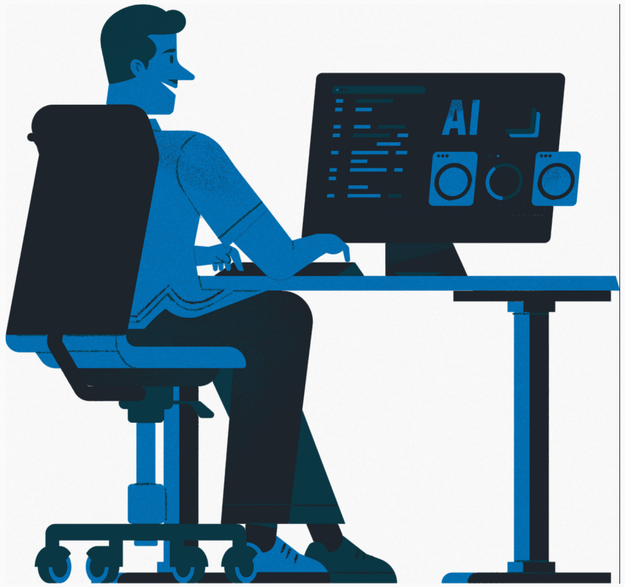
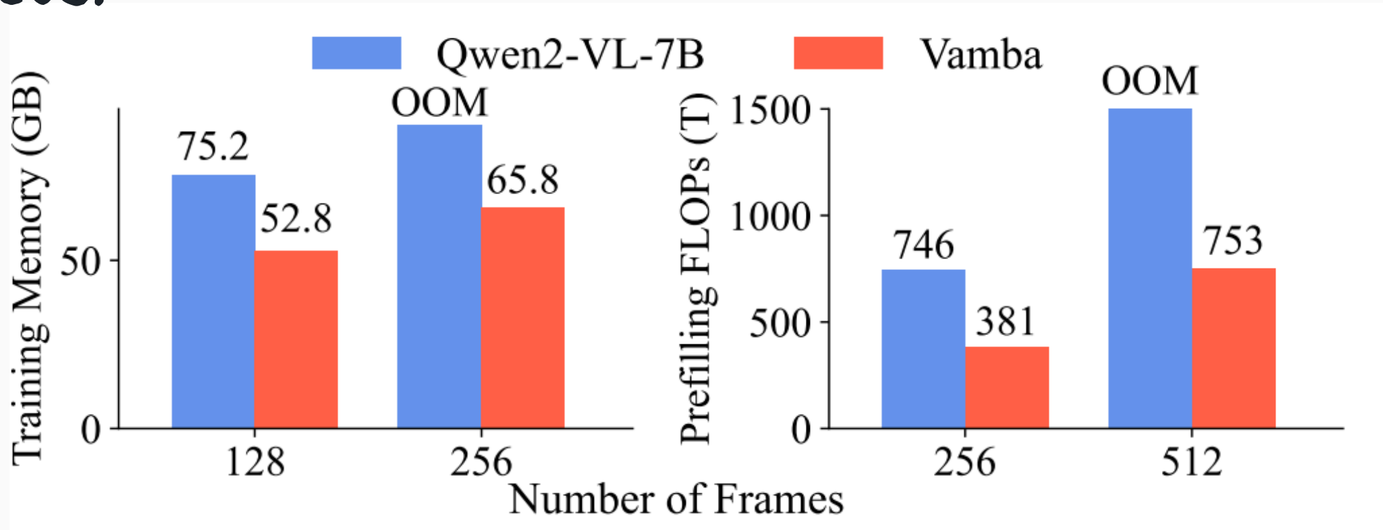
Today's Flow



- Why long video understanding is difficult
- Existing solutions and their problems
- VAMBA: New hybrid architecture
- Experimental results
- Efficiency gains
- Case studies and key takeaways

Problem with Long Videos

- Transformers have quadratic complexity.
- Huge GPU memory + time costs.
- Only 256 frames processed — not enough for 1-hour videos.

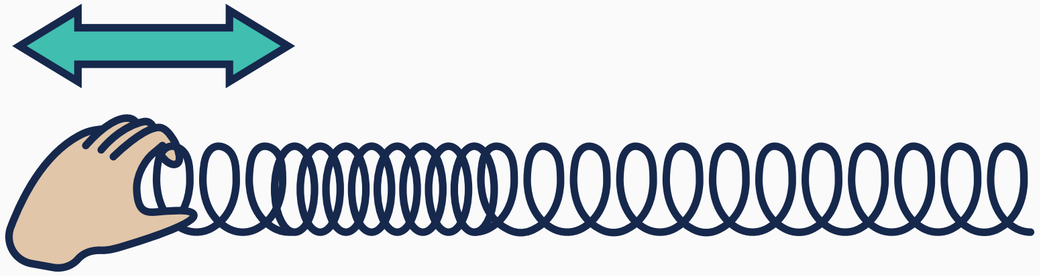


How Prior Models Tried to Solve It

Method	How it works	Problems
Q-Former	Compresses visual tokens into fewer tokens	Loses fine-grained details
LongVU	Adaptive chunk compression of vision tokens	Still quadratic internally; aggressive compression harms
Video-XL	Summarizes visual frames into fewer summary tokens	Loss of temporal dynamics (time-sensitive events)
InternVideo 2.5	Drops less important video frames dynamically	Important events can be missed; runtime still heavy

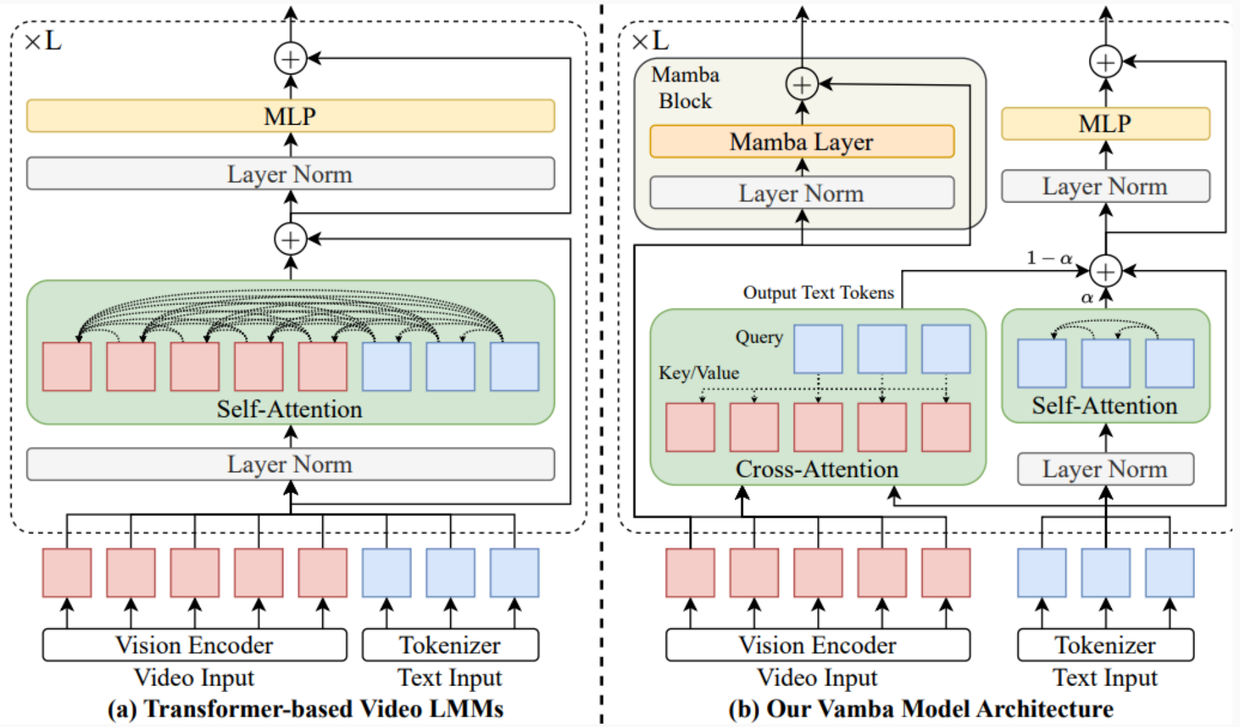
A Different Approach: No Compression!

- No token dropping.
- Handle 1000+ frames without losing information.
- Linear complexity achieved by architecture change.



How VAMBA Works Internally

- Text tokens: Cross-Attention layers.
- Video tokens: Mamba-2 Blocks (linear-time SSM).
- Combined for efficient multimodal understanding.



Efficient Text Token Updating

- Instead of attending to everything, text tokens focus on video context.
- Cross-attention + small self-attention.
- Reduces computation dramatically.

$$o_{t_j} = (\sigma(\frac{q_{t_j}[\mathbf{K}_v, \mathbf{K}_{[t_1:t_j]}]^\top}{\sqrt{d}})[\mathbf{V}_v, \mathbf{V}_{[t_1:t_j]}])\mathbf{W}_o, \quad (2)$$

Efficient Video Token Updates

- Mamba-2 processes sequential tokens efficiently.
- Maintains dependencies without full attention computation.
- $O(d^2M)$ instead of $O(M^2)$.

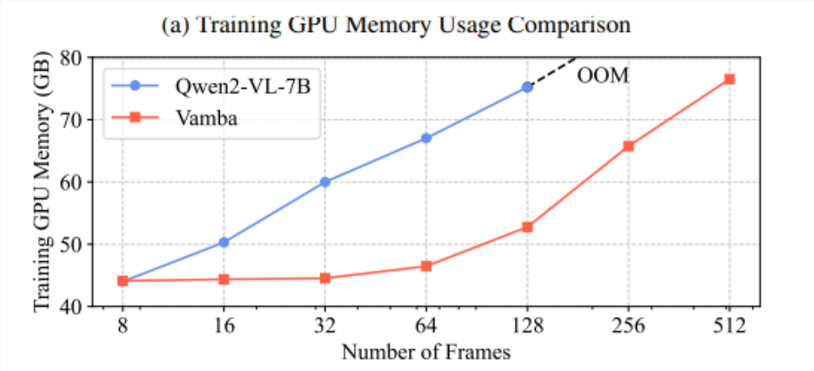
Performance Benchmarks

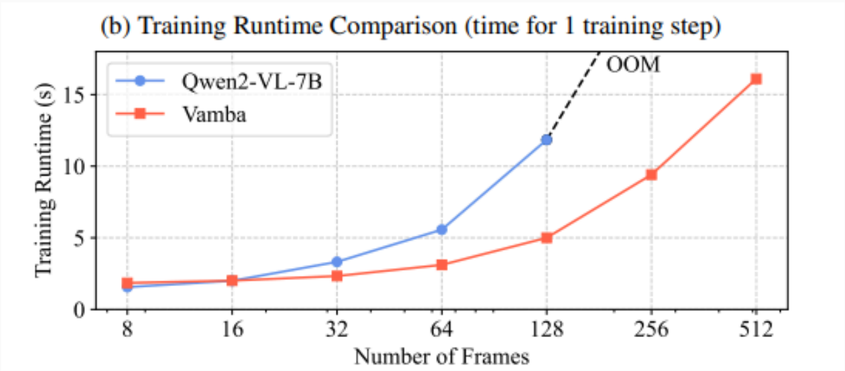
- LVBench: 42.1% accuracy (best among efficient models).
- 50% memory reduction during training.
- 2× training speedup over transformer baselines.

Models	Size	Hour-Long Video Understanding		
		LVBench	HourVideo	HourEval
		test	dev	test
Transformer-based LMMs				
Gemini-1.5-Pro [58]	-	33.1	37.3	-
GPT-4o [49]	-	34.7	19.6	-
Qwen2-VL [62]	7B	42.0	33.8	53.0
Efficient LMMs				
LLaVA-Mini [74]	7B	17.6	20.6	24.2
LongLLaVA [64]	9B	31.2	27.7	39.1
LongVU [54]	7B	37.8	30.8	46.8
Video-XL [55]	7B	36.8	33.0	47.1
VAMBA	10B	**42.1**	**33.6**	**50.7**

Memory and Runtime Gains

- Trains larger videos (512 frames vs 128 frames).
- Pre-fill FLOPs cut by 30–50%.
- Full training on 8 GPUs instead of 64 GPUs!



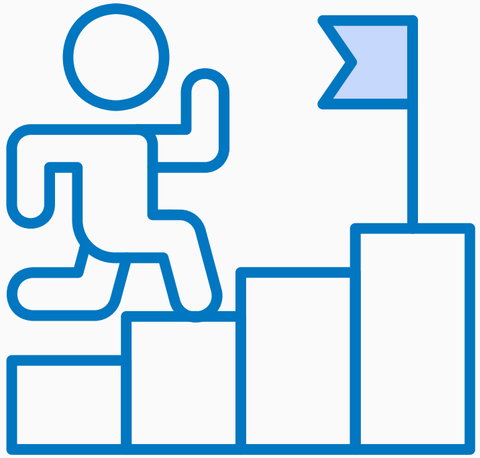


VAMBA vs Video-XL: Real-World Examples

- Cookie video: VAMBA correctly identified cookie colors.
- Magic trick: VAMBA accurately described actions with fine detail.

What Helped VAMBA Perform Better?

- Initializing Cross-Attention weights improved performance.
- Mamba-2 blocks worked better than simple Mamba.
- Distillation loss didn't help — dropped.



VAMBA vs Other Efficient Models

Feature	LongVU	Video-XL	Vamba
Max Frames	256	256	1024
GPU Memory Usage	High	High	Low
Accuracy (LVBench)	37.8%	36.8%	42.1%
Information Loss	Yes	Yes	No

Key Takeaways

- VAMBA proves you can process hour-long videos efficiently.
- Hybrid attention architectures = future trend.
- Next step: Combine VAMBA + smart token reduction.

THANK YOU

Any Questions?